



Autumn Examinations 2013 / 2014

Exam Code(s) 1SD1/1SD3/1MF1/1IT1
Exam(s) Higher Diploma in Applied Science (Software Design & Development), Higher Diploma in Applied Science (Software Design & Development) Industry Stream, M.Sc. in Software Design & Development, Master of Information Technology

Module Code(s) CT874
Module(s) Programming I

Paper No. I

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Instructions: Candidates are required to answer:

Section A (20 Marks) Answer all 10 parts of section A, This section consists of 10 multiple-choice questions all of which should be attempted. **Answers are to be written on the MCQ sheet provided, NOT on the Examination paper.**

Section B (80 Marks) Answer Any **Two** Questions.

Duration 2 hours
No. of Pages 6
Department(s) Information Technology

Requirements MCQ

Section A (20 Marks)
Attempt all 10 questions
Mark your answers on the MCQ sheet provided

- 1) The practice of writing multiple constructors to handle different sets of inputs is known as _____.
 - a) inheritance
 - b) polyinputting
 - c) overloading
 - d) multiplicity

- 2) When an exception occurs, the normal sequence of flow is terminated and the exception is _____.
 - a) accessed
 - b) thrown
 - c) terminated
 - d) initialized

- 3) Which of the following is not a Boolean operator?
 - a) &&
 - b) !
 - c) ||
 - d) ++

- 4) Polymorphism allows us to _____.
 - a) have the same message execute a different method depending on the receiving object
 - b) hide the internal workings of a Java program from the client programmer
 - c) overload methods and allow the same method name to be used multiple times
 - d) extend and modify code

- 5) Which of the following declares an array of integers of size 20?
 - a) `int[] a = int[20];`
 - b) `int[] a = new int[20];`
 - c) `int a = new int[19];`
 - d) `int a[] = new int[19];`

6) Which of the following methods is not contained within the List class?

- a) hasNext
- b) add
- c) List
- d) remove

7) The String class method, toUpperCase _____.

- a) Makes lowercase characters in the string uppercase and uppercase characters lowercase.
- b) Makes the first character of the string uppercase and the rest lowercase.
- c) Converts a single character to uppercase.
- d) Converts the entire String to uppercase.

8) In computer science, what is meant by garbage collection?

- a) Deallocation of unused memory space.
- b) Going through the program and fixing erroneous code.
- c) Deletion of objects from memory.
- d) Defragmentation of the disk drive.

9) What is meant by “returning an object” from a method?

- a) It means you are returning a null value to the caller.
- b) It means returning a primitive type to the caller.
- c) It means returning the object’s address to the caller.
- d) It means returning the object’s value to the caller.

10) What does the call, “super()” do?

- a) It will do nothing.
- b) Call the subclass’s constructor
- c) Call the superclass’s constructor.
- d) This is an invalid call.

Section B (80 Marks)
Answer Any TWO Questions

Q 2.

- a) In relation to object-oriented programming with Java, detail your understanding, using examples/analogies where appropriate, of each of the following:

- i. Polymorphism,
- ii. Interface
- iii. Class member
- iv. Overriding

[10]

- b) Write a Java application, which allows the user to input a line of text. The application should output the line of text in both uppercase and lowercase.

[15]

- c) Create a Java application that takes as input a string and uses a loop to count the number of uppercase characters contained in the string.

[15]

Q 3.

- a) Explain the terms **composition** and **inheritance** as used in object-oriented programming. You should provide snippets of Java code to illustrate your answer.

[10]

- b) Develop a Java application which will be used to calculate the interest for a savings account.

You are required to:

- i. Create a **Savings** class which includes a static variable *interestRate* to hold the annual amount of interest for each account holder. The Savings class contains the following data members:

- *balance* (double)
- *interestRate* (double)

The *balance* data member holds the current balance of the account, while the *interestRate* data member holds the current rate of interest. Both use appropriate getter and setter methods.

[10]

You are also required to create a **calculateInterest** method, which calculates the interest earned by multiplying the balance by the rate of interest. This figure should then be added to the balance.

[5]

Also create a static **changeInterestRate** method that allows the *interestRate* to be changed to a new value.

[5]

- ii. Write a **client class** to test the *Savings* class and **display** the balance, and interest earned for each account holder.

[10]

Q 4.

Irish Sales Ltd. require a payroll application written in Java. The workforce is made up of two different types of employees, an hourly employee, who is paid by the hour and a commission employee, who is paid only on a commission basis.

a) Create a **Employee** class which has the following data members:

- firstName (static)
- latName (String)
- ppsNumber (String)

These data members are tracked using appropriate **setter** and **getter** methods. Hourly employees are paid €15.00 per hour and commission based employees are paid 10% of the value of their sales figure.

You are also required to appropriately plan for a **earnings** method, which will be overridden –

double earnings();

[15]

b) You are also required to create a **HourlyEmployee** class and a **CommissionEmployee** class that are *derived* from Employee. In each of these classes, create an overwritten method named **earnings**. Note: The *HourlyEmployee* class and the *CommissionEmployee* should use appropriate **setter** and **getter** methods.

[15]

c) You are required to test your classes using a **client class** and to store the instances of employees into an Array.

[10]